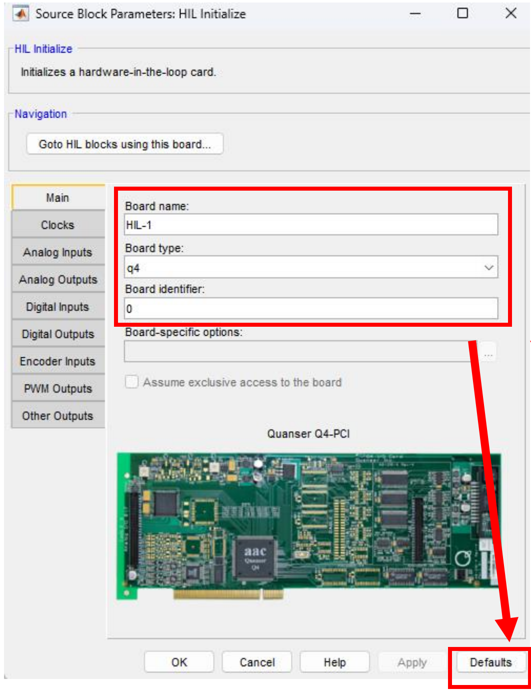
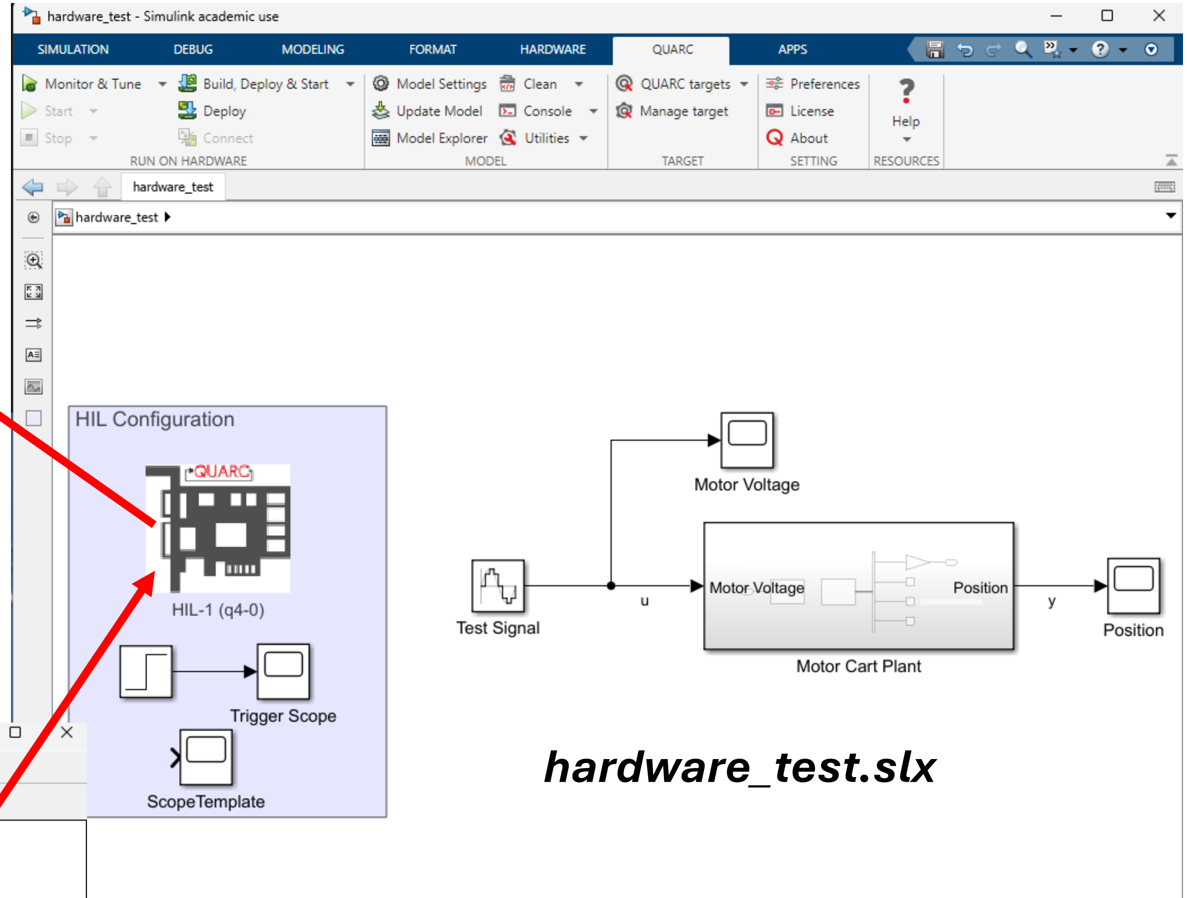
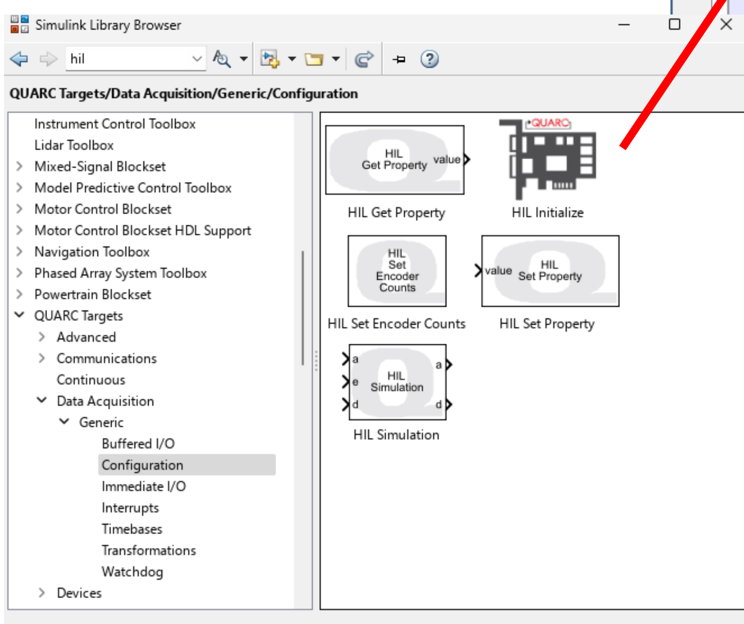
Configuring the model to work with DAQ

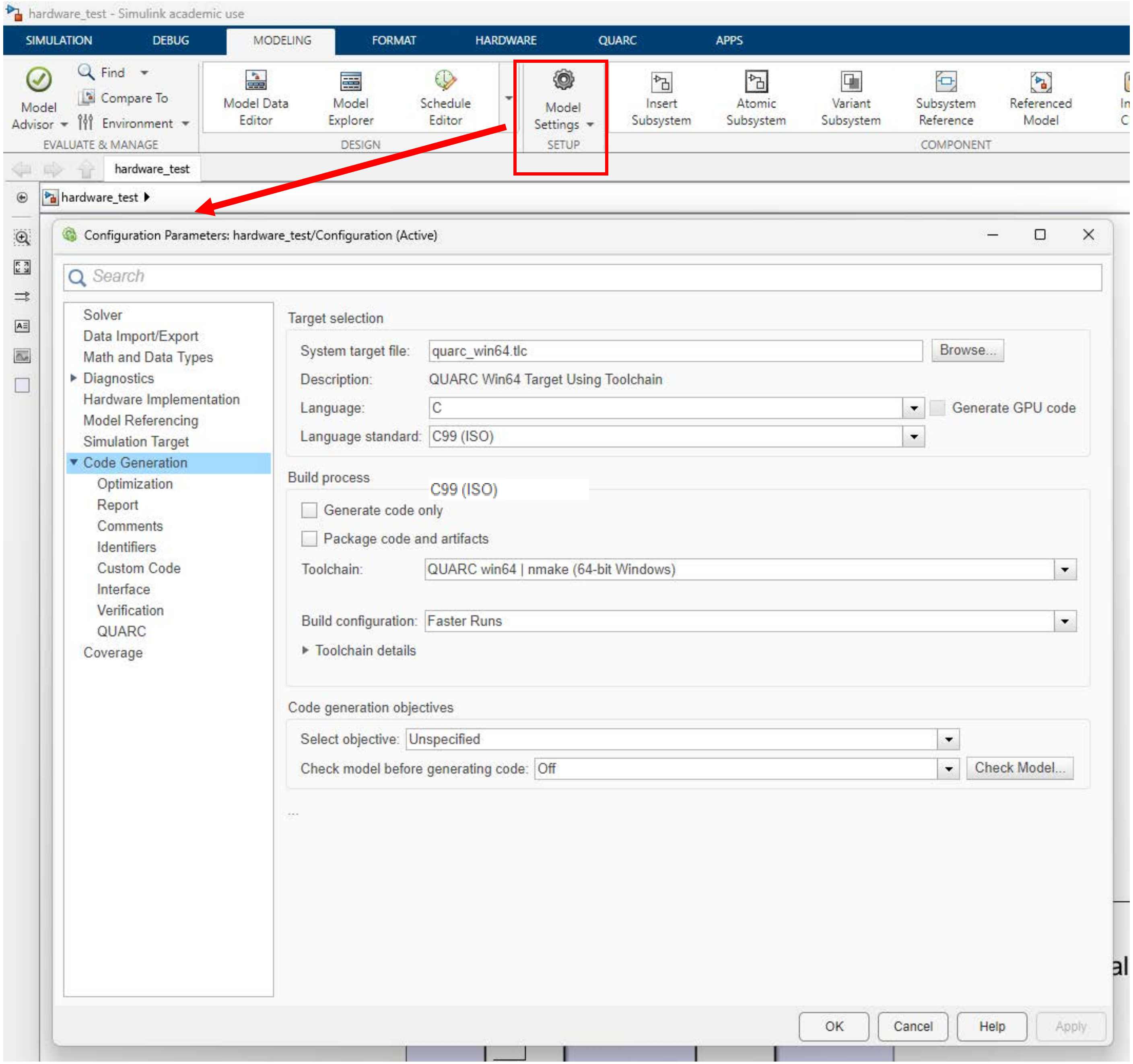


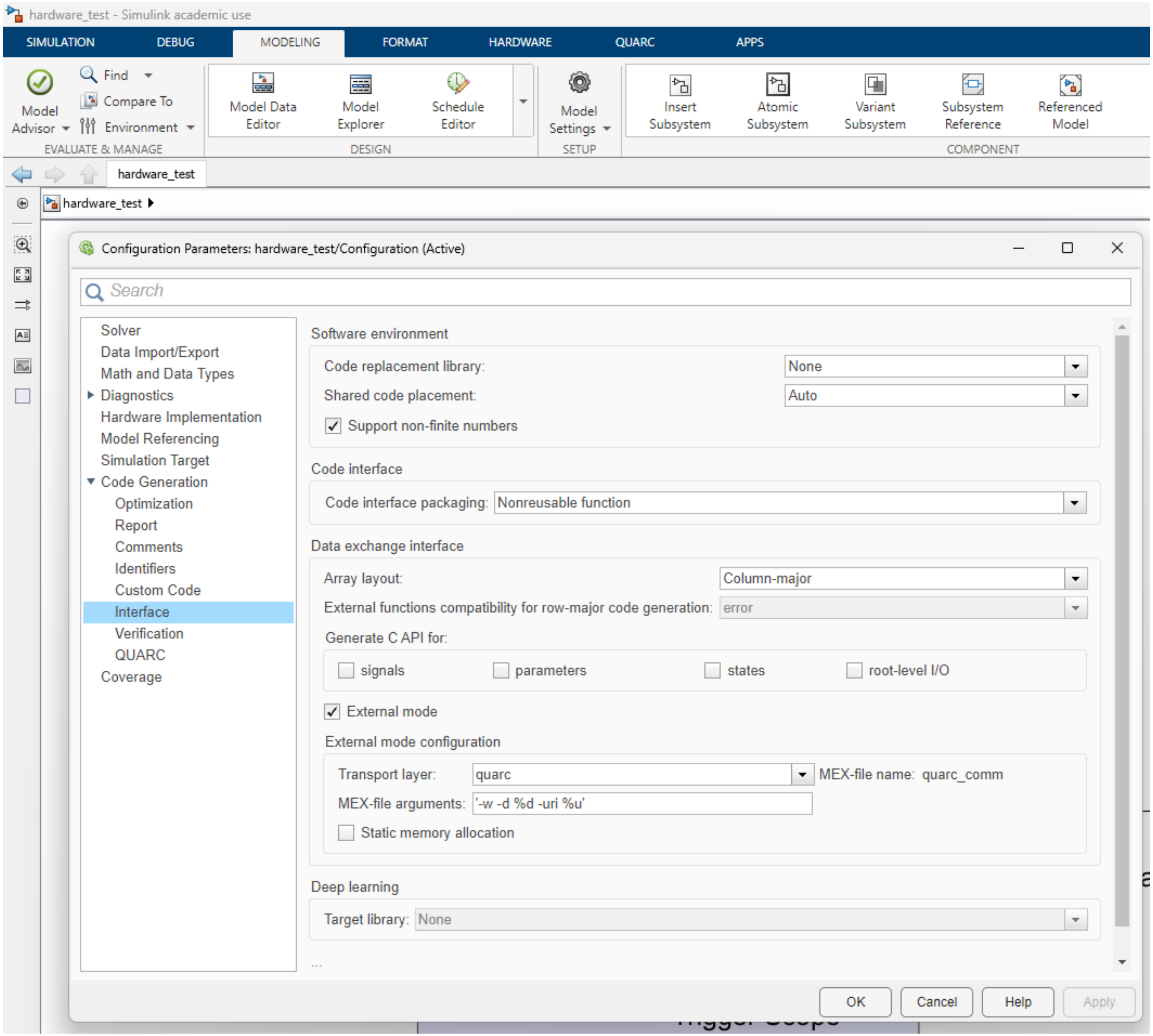




Set the HIL Initialize block to have board name HIL-1 and Board type q4. Click defaults to set up

Configuring model parameters for HIL & Quarc




hardware_test - Simulink academic use
SIMULATION
DEBUG
MODELING
FORMAT
HARDWARE
QUARC
APPS
Model Advisor
Find
Compare To
Environment
EVALUATE & MANAGE
Model Data Editor
Model Explorer
Schedule Editor
DESIGN
Model Settings
SETUP
Insert Subsystem
Atomic Subsystem
Variant Subsystem
Subsystem Reference
Referenced Model
COMPONENT
hardware_test
hardware_test
Configuration Parameters: hardware_test/Configuration (Active)
Search
Solver
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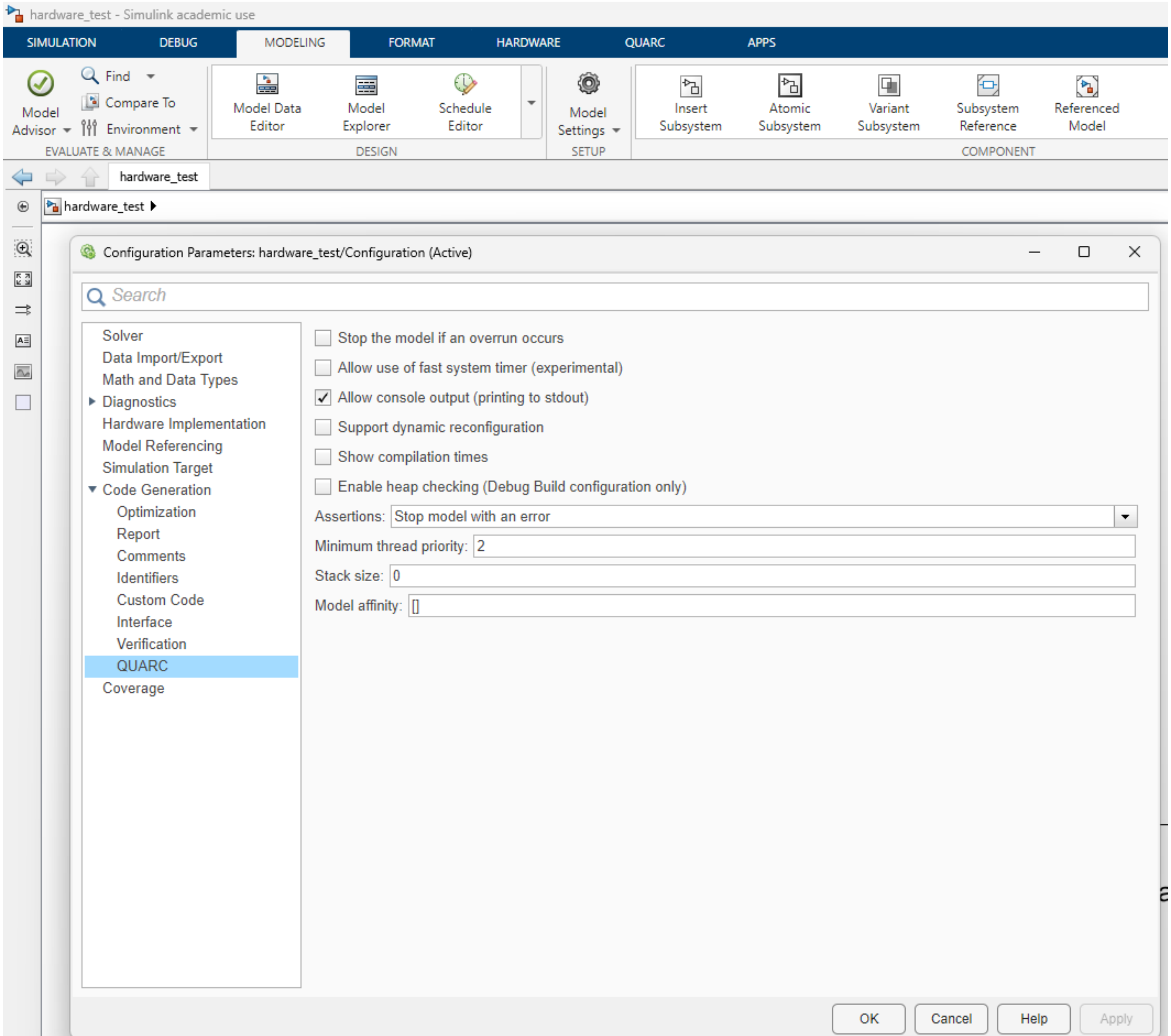
Math and Data Types
Diagnostics
Hardware Implementation
Model Referencing
Simulation Target
Code Generation
Optimization
Report
Comments
Identifiers
Custom Code
Interface
Verification
QUARC
Coverage
Software environment
Code replacement library: None
Shared code placement: Auto
Support non-finite numbers
Code interface
Code interface packaging: Nonreusable function
Data exchange interface
Array layout: Column-major
External functions compatibility for row-major code generation: error
Generate C API for:
signals
parameters
states
root-level I/O
External mode
External mode configuration
Transport layer: quarc
MEX-file name: quarc_comm
MEX-file arguments: '-w -d %d -uri %u'
Static memory allocation
Deep learning
Target library: None
OK
Cancel
Help
Apply


hardware_test - Simulink academic use
SIMULATION
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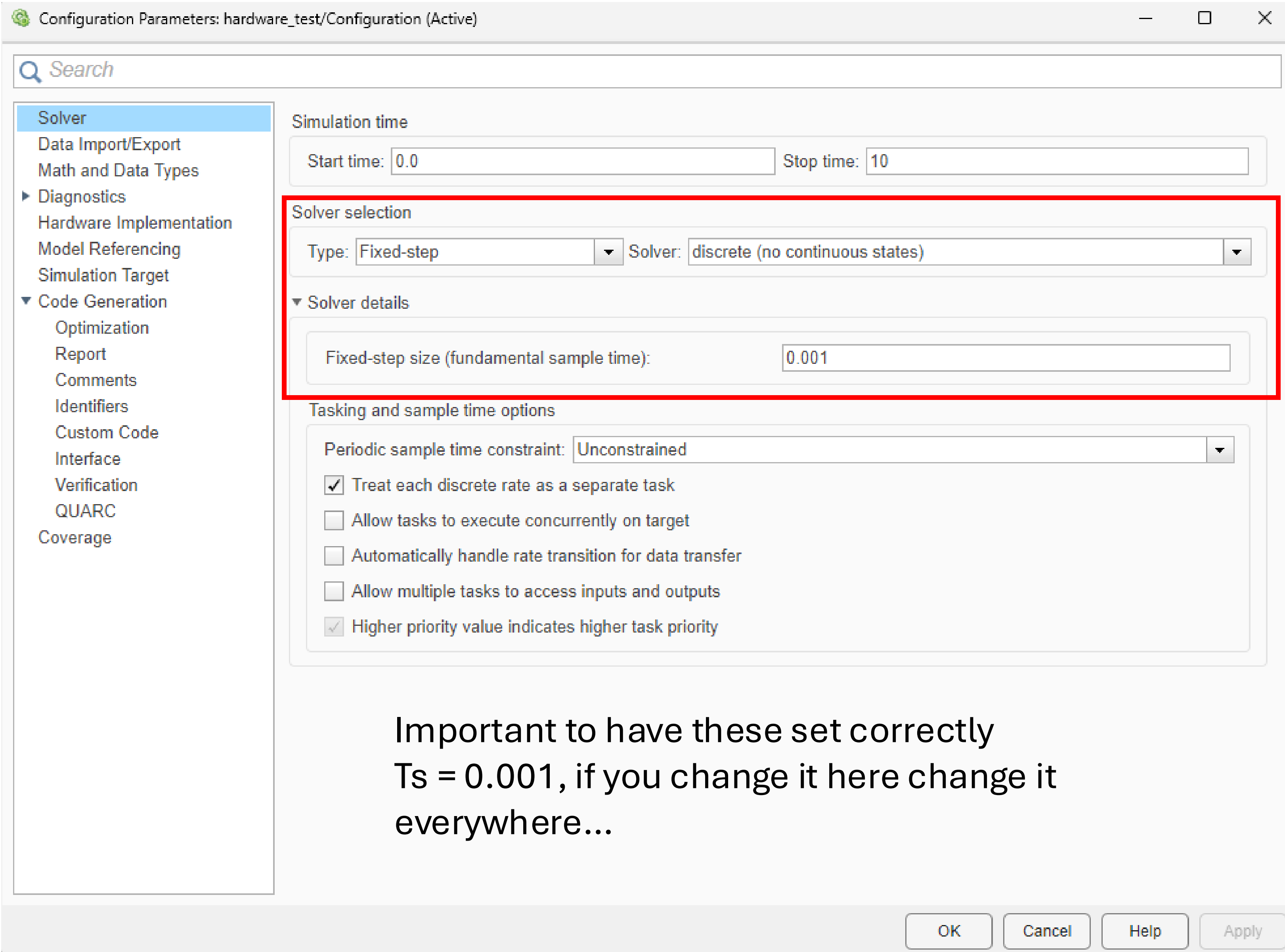
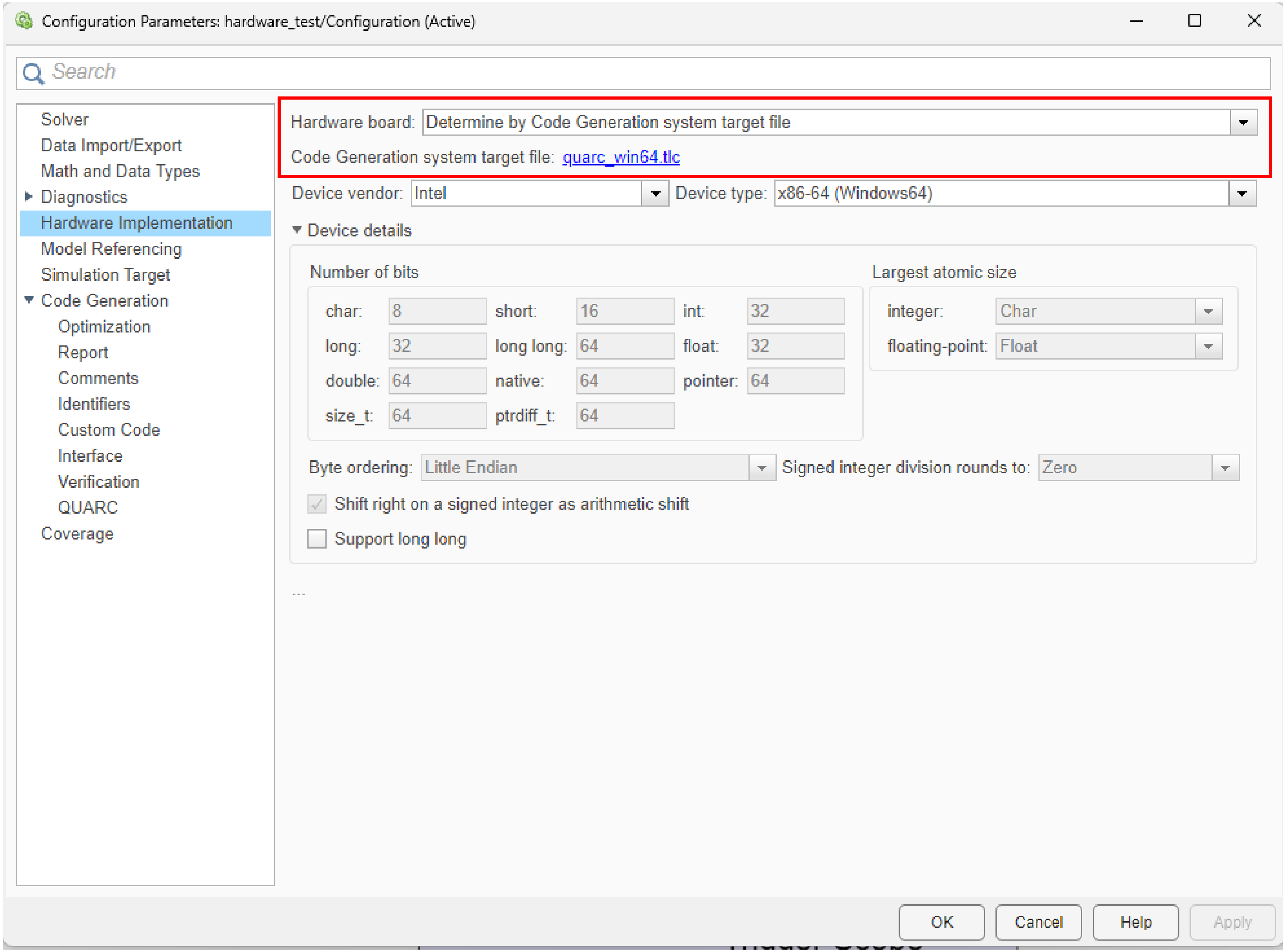
Custom Code
Interface
Verification
QUARC
Coverage
Stop the model if an overrun occurs
Allow use of fast system timer (experimental)
Allow console output (printing to stdout)
Support dynamic reconfiguration
Show compilation times
Enable heap checking (Debug Build configuration only)
Assertions: Stop model with an error
Minimum thread priority: 2
Stack size: 0
Model affinity: []
OK
Cancel
Help
Apply

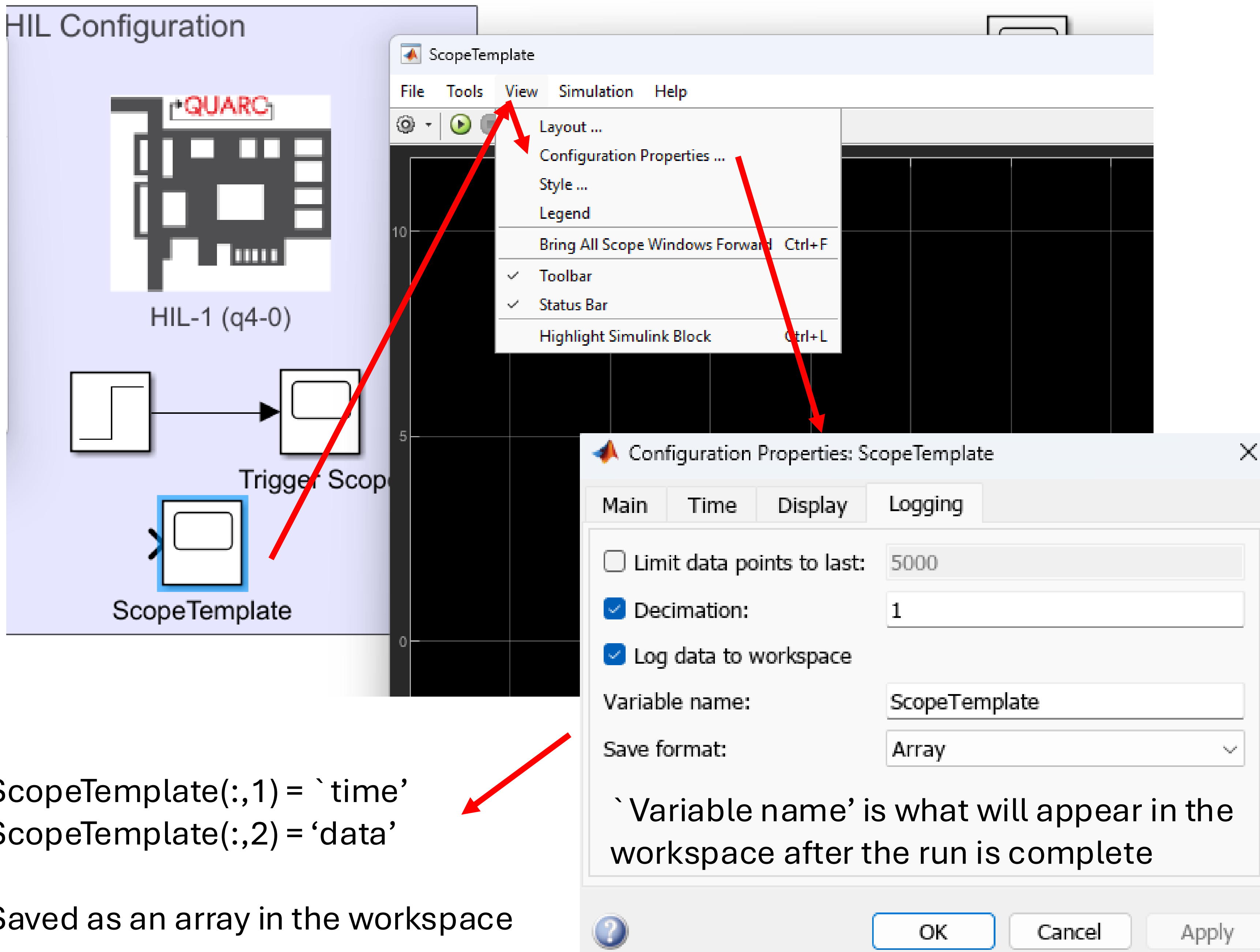


Important to have these set correctly
Ts = 0.001, if you change it here change it everywhere…



Configuration Parameters: hardware_test/Configuration (Active)
Search
Solver
Data Import/Export
Math and Data Types
Diagnostics
Hardware Implementation
Model Referencing
Simulation Target
Code Generation
Optimization
Report
Comments
Identifiers
Custom Code
Interface
Verification
QUARC
Coverage
Hardware board: Determine by Code Generation system target file
Code Generation system target file: quarc_win64.tlc
Device vendor: Intel
Device type: x86-64 (Windows64)
Device details
Number of bits
char: 8
short: 16
int: 32
long: 32
long long: 64
float: 32
double: 64
native: 64
pointer: 64
size_t: 64
ptrdiff_t: 64
Largest atomic size
integer: Char
floating-point: Float
Byte ordering: Little Endian
Signed integer division rounds to: Zero
Shift right on a signed integer as arithmetic shift
Support long long
OK
Cancel
Help
Apply

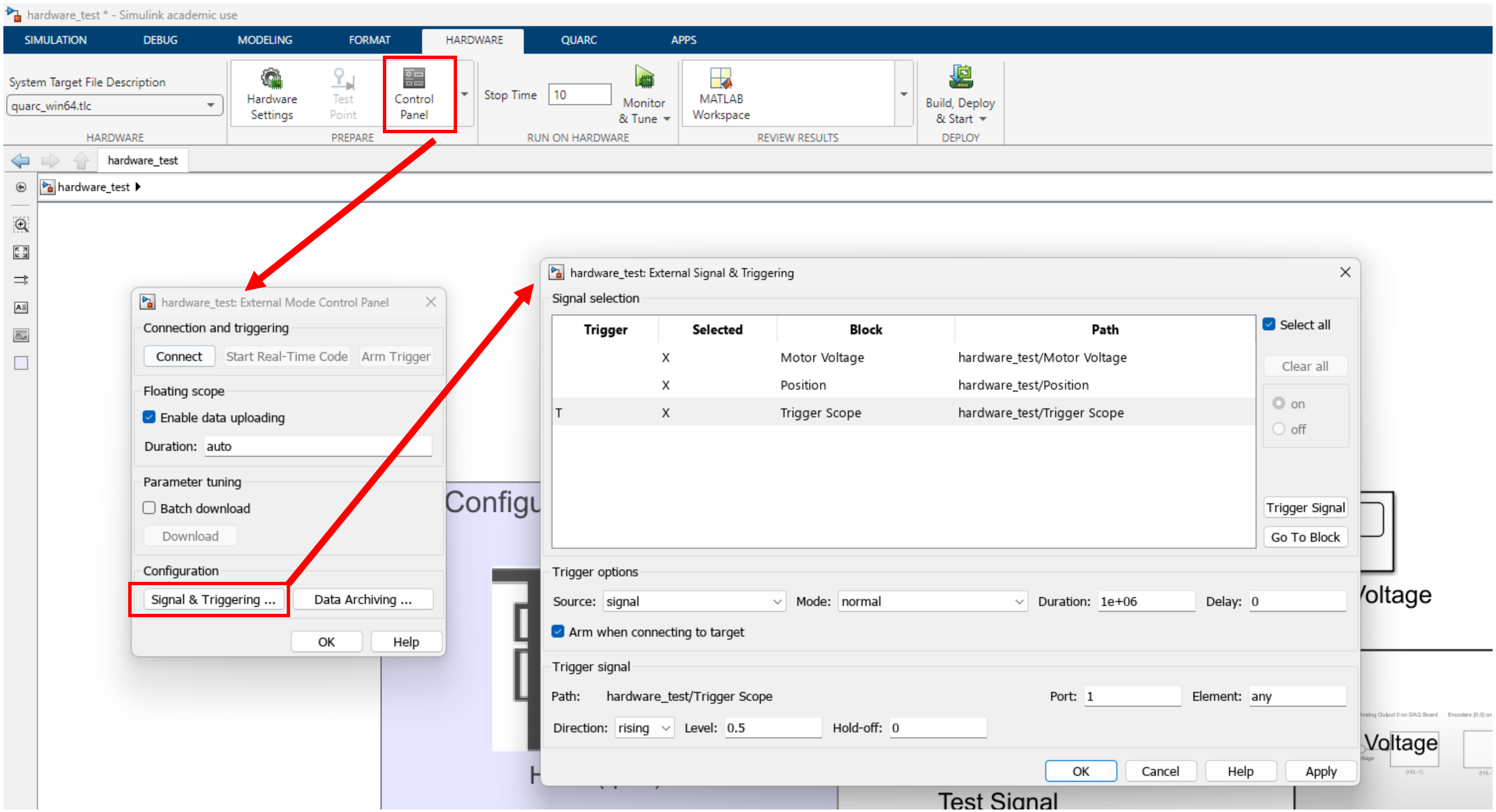
Configuring scopes for data logging:



ScopeTemplate(:,1) = `time'
ScopeTemplate(:,2) = 'data'

Saved as an array in the workspace

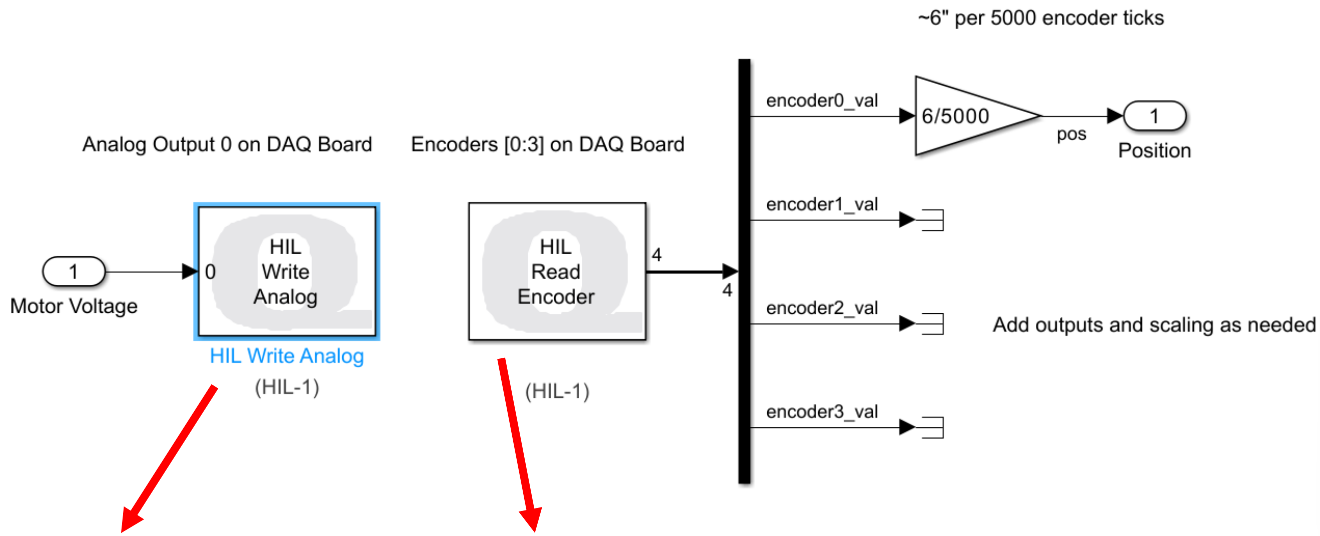
Configuring triggering and model data logging:

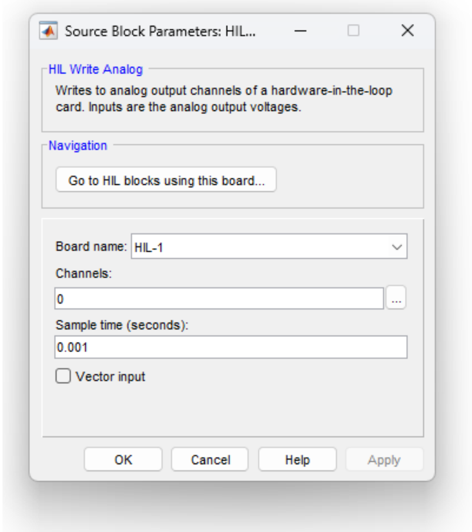


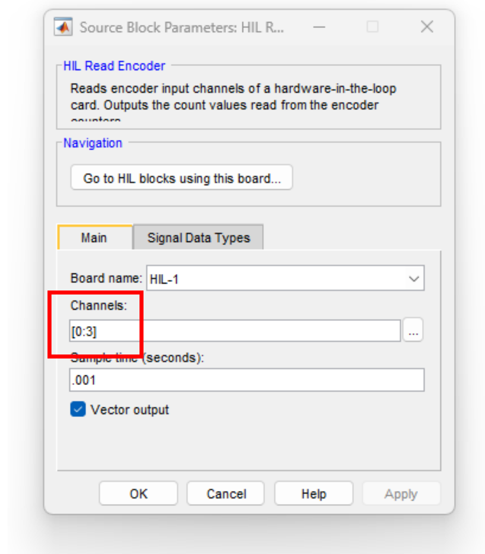
X under Selected means the signal will be logged

T under trigger means that will be used as trigger signal (don’t change)

Duration is maximum # of samples logged, *Duration = tStop/Ts*

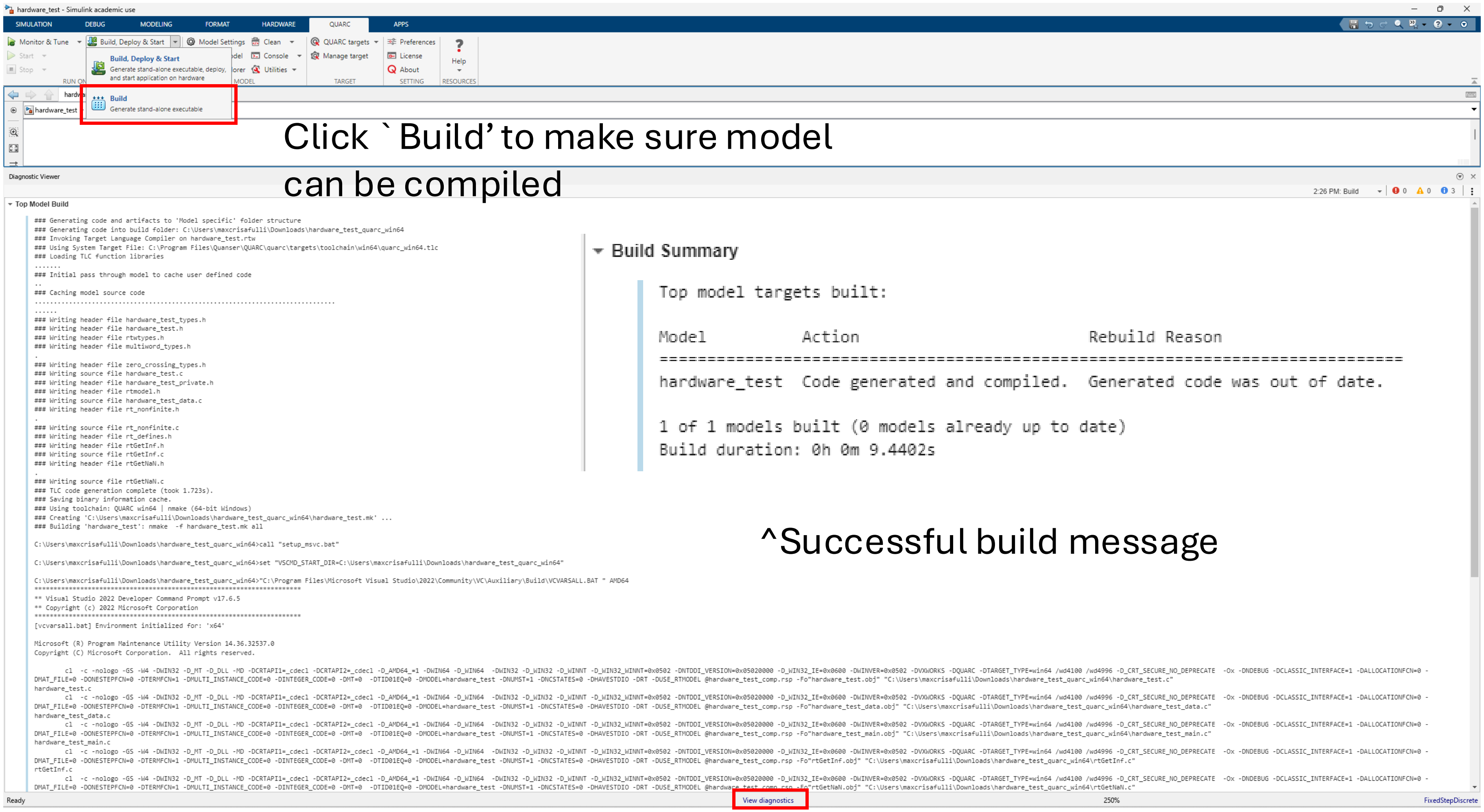






Demo model is reading all 4 encoders as vector and demuxxing (shown above)

Can set to just read a single encode



Click ` Build' to make sure model can be compiled

^Successful build message

Monitor & Tune builds & runs the code.
Expected outputs shown